

The data for this mini project comes from basketball-reference.com, a popular website that lists NBA statistics for players and teams across the history of the league. The data I used for this project consists of player and team statistics from the 2024-25 season, which just ended. The goal with this project is to show off the best players of the season as well as to visualize the environment of the league through team statistics and summary statistics of player stats.

Basketball Reference has a very clean and organized data set, however I decided to take an extra step to clean the data by removing players who played less than 10 minutes per game, just to get rid of any outliers that could affect summary stats due to small sample size bias. Certain players spent this season on more than one team, and Basketball Reference handles this by listing player statistics in individual rows by the team they played for, as well as a final row that shows their overall season statistics across all teams. For example, Luka Doncic was traded from the Dallas Mavericks to the Los Angeles Lakers in the middle of the season, so he has three rows in this data set, one for Mavericks statistics, one for Lakers statistics, and one for his entire season. Because I would go on to merge my player and team data sets, I removed every “entire season” row so players would only show statistics they put up for the team represented in their row. Every row after merging my data contains a player’s statistics, as well as the win totals and points scored/allowed per game stats of the team they played for.

My program analyzes the full data set in multiple different ways. Firstly, it produces a table of league leaders in four major stat categories: points, rebounds, assists, and field goal percentage, showing the top ten players in each category. It does the same with team statistics, showing the top ten teams in points scored per game and points allowed per game. It also allows

the user to select any team and generate a table with all the requisite statistics for an individual team, while saving the table as a CSV under the team's name. Finally, two subsets of the initial data set were created with one table showing statistics only for each team's five most started players, and the other table showing all other players. Summary statistics were shown for starters and non-starters, with the result seeming to be that starters score the ball more efficiently.